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Projectile Motion of a Marble Shooter

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Abstract: The objective of this lab is to analyze the relationship between launch angle and horizontal range of objects in projectile motion through demonstrations of a marble being shot by a spring-loaded shooter. A marble having a particular mass (g) was launched from the spring-loaded shooter at 12 different launch angles ranging from 0° to 82.5°, and five trials were conducted for each of the launch angles. It was found that there was a parabolic relationship between the launch angles and horizontal range, with the optimum value in terms of distance was about 45°. Quadratic regression produced the result that $y = -0.031506x^2 + 2.64198x + 7.26186$ with $R^2 = 0.9961$.

I. Introduction:

Projectile motion occurs when an object travels through a two-dimensional plane under the influence of gravity, while the horizontal motion has a constant velocity ($v_0x = v_0\cos\theta$) and vertical motion under constant downward acceleration ($v_0y = v_0\sin\theta - gt$). For a launch speed v_0, launch angle θ, and launch height y_0, the position at time t is:

$$x(t) = (v0\cos\theta)t \quad (1)$$

$$y(t) = y0 + (v0\sin\theta)t - \frac{1}{2}gt^2 \quad (2)$$

Combining the horizontal position function and the vertical displacement function yields the general parabolic trajectory equation seen below:

$$y(x) = y0 + x\tan\theta - \left(\frac{g}{2v_0^2\cos^2\theta}\right)x^2 \quad (3)$$

For launches and landings on level ground, setting the vertical displacement to zero yields the flight time $t = \frac{2v_0\sin\theta}{g}$, and then using a double-angle identity you find the theoretical range:

$$R = \frac{v_0^2\sin(2\theta)}{g} \quad (4)$$

Equation (4) predicts that the farthest horizontal distance should occur at approximately 45°, and that two complementary angles (θ and $90° - \theta$) give the same range. When the projectile is launched from a height y_0 above the landing surface, the range becomes:

$$R = \frac{v_0\cos\theta}{g}\left(v_0\sin\theta + \sqrt{v_0^2\sin^2\theta + 2gy_0}\right) \quad (5)$$

and the maximum moves below 45°. This formulaic setup is the base for checking the experimental path and measured range values for all the different launch angles that were tested.

The shape of the path was not, historically, always understood. Tartaglia (1537) described a projectile moving along a straight line, then a circular arc, then falling vertically. Galileo (1638) showed that the horizontal and vertical motions are independent and that the path is a parabola. Our shooter records only where the marble lands, so we tested the two quantities that equations (4) and (5) predict from landing points: the angle of maximum range and the symmetry of range about 45°.

Hypothesis 1 (angle of maximum range):

$$H_0: \theta_{max} = 45^\circ \qquad H_1: \theta_{max} \neq 45^\circ$$

Hypothesis 2 (symmetry): For each complementary pair of angles,

$$H_0: \bar{R}(\theta) = \bar{R}(90^\circ - \theta) \quad H_1: \bar{R}(\theta) \neq \bar{R}(90^\circ - \theta)$$

where $\bar{R}$ is the mean range. H_0 in both cases is the ideal level-ground prediction of equation (4), and we expected to fail to reject it because the shooter fires from close to the landing surface. Rejecting H_0 would mean launch height, air resistance, or equipment error measurably changes the range.

Materials/Methods:

A. Setup

In our lab, a marble shooter, base plate, clamp, protractor, measuring tape, nine marbles, a balance, white paper, and blue carbon/graphite paper were employed. At the start, the base plate was set exactly at the edge of the table, pointing away from anyone. It was then clamped firmly so that the position is constant throughout the whole experiment. Then, the marble shooter was fixed to the right side of the base plate using a bolt, washer, and nut. We also make sure to firmly screw the nut by using pliers to stabilize the shooter. We also made sure a pencil that was fixed to the right side of the shooter's string as the weight to properly show the angle of the protractor.

Then, we measured the horizontal distance from the edge of the base plate to the edge of the table (inches) and noted it. Moreover, the total mass of nine marbles was measured using a balance, and the total mass was divided by nine to find the average mass of one marble (grams). During each trial, a marble was placed inside the shooter and set at the first, weakest, setting. The marble shooter was then rotated until the angle pointed at the designated angle on the protractor, making sure the shooter was stable and the launching path was clear.

B. Launch tests

For each trial, using the same starting point, a marble was loaded at the first (weakest) spring setting, and the shooter was rotated to the target angle on the protractor. The string was pulled back fully and released. After the first shot at each angle, we located the landing spot, laid

white paper with carbon paper on top, and repeated the shot so the marble left a blue mark on the paper. We measured the distance from the same reference point on the base plate to the mark (inches). Five trials were conducted at each of 12 different angles (0°, 7.5°, 15°, 22.5°, 30°, 37.5°, 45°, 52.5°, 60°, 67.5°, 75°, 82.5°), for 60 trials total. The shooter was returned to the same position before each angle, and we checked the data table for five recorded distances per angle before taking the setup apart. Then, the marbles and paper were then put away, and disassembly of the marble shooter was done by taking off the nut, washer, and bolt, and then the clamp and base plate. All the equipment was put back in their respective places, and cleaning was done of the experimentation area.

C. Analysis

We computed the mean and sample standard deviation (n = 5) of the distance traveled at each angle. Using Desmos, we fit a quadratic regression to all 60 points (distance vs. launch angle) and recorded the equation and R^2. Then, we estimated the angle of maximum range from the vertex of the fitted parabola, $x = -b/(2a)$. To test hypothesis 1 (maximum at 45°), we compared the mean distance at 45° with the mean distances at 30°, 37.5°, and 52.5° using two-sided two-sample t-tests ($\alpha = 0.05$). Furthermore, to test hypothesis 2 (symmetry), we used the same test on each of the complementary pairs of angles: 7.5° and 82.5°, 15° and 75°, 22.5° and 67.5°, 30° and 60°, and 37.5° and 52.5°.

Results:

The distance traveled (inches) by each marble with respect to its angle is shown in Table 1. There were 5 trials conducted for each angle. Graph 1 shows the projectile motion trajectory graph with all the angles and marble distance traveled plotted.

Table 1.

#	Angle (°)	#	Trial 1 distance traveled (in)	Trial 2 distance traveled (in)	#	Trial 3 distance traveled (in)	#	Trial 4 distance traveled (in)	#	Trial 5 distance traveled (in)	Mean distance traveled ± s.d. (in)
	0		9	9.5		9		9.5		7.5	8.90 ± 0.82
	7.5		24.25	24.15		24.81		24.44		22.88	24.11 ± 0.73
	15		38.31	38.43		38.5		38.75		38.38	38.47 ± 0.17
	22.5		50.31	50.06		50.13		50.44		50.31	50.25 ± 0.15
	30		60.81	58		57.56		57.38		57.93	58.34 ± 1.41
	37.5		62.5	62.44		62.31		62.38		62.25	62.38 ± 0.10
	45		62.81	63.38		62.81		62.38		62.56	62.79 ± 0.38
	52.5		62.7	60.75		60		60.38		60.56	60.88 ± 1.06
	60		52.63	52.13		52.88		52.69		51.5	52.37 ± 0.56
	67.5		41.81	40		40.06		40.81		40.69	40.67 ± 0.73
	75		27.75	26.81		27.81		27.5		27.88	27.55 ± 0.44
	82.5		12	10.81		11.13		11.56		11.44	11.39 ± 0.45

https://docs.google.com/spreadsheets/d/17ZYdJSj7SxPZqcI_80ieY-MBuT8ozDmQGkfJQnoRS68/edit?usp=sharing

Graph 1.

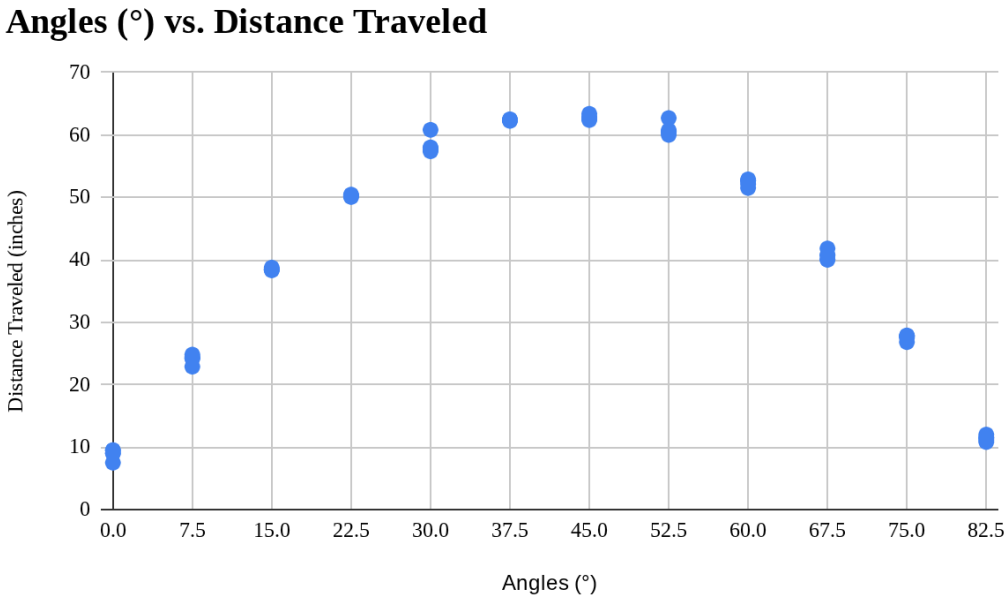


Figure 1: The resulting trajectory is shown in the graph above, yielding an equation of $y = -0.031506x^2 + 2.64198x + 7.26186$ with an R^2 of 0.9961. Each point on the graph represents distance traveled by the marble at that specific angle. All the measurements were plotted in Desmos and a quadratic regression was run to find a mathematical equation that best fitted those points to replica real-world data.

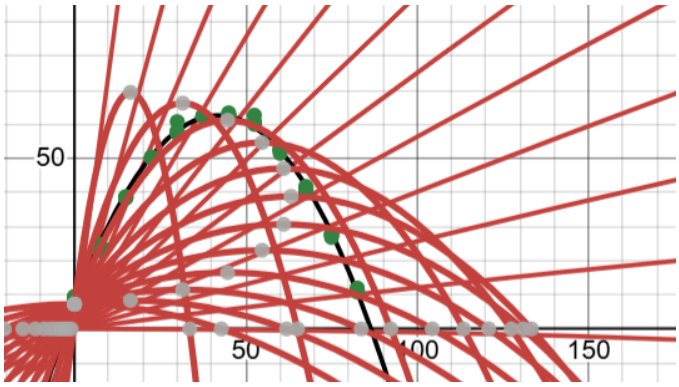


Figure 2: Galileo's graph compared with Figure 1 data.

IV. Discussion:

Tartaglia vs. Galileo

Tartaglia's hypothesis with regard to projective motion states that objects move in a curved arc and then drop straight to the ground. Comparatively, Galileo's hypothesis argued that a projectile moves in a parabolic path, similar to the data collected and displayed in Graph 1.

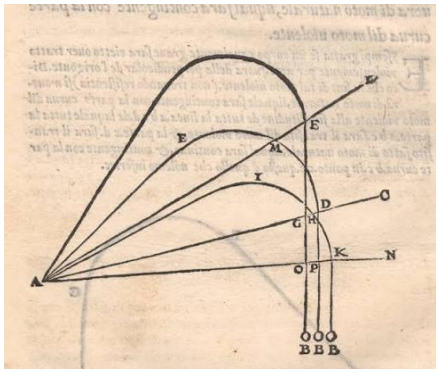


Figure 2: Tartaglia's projectile motion graph. It is a qualitative conceptual framework.
Image from EOS

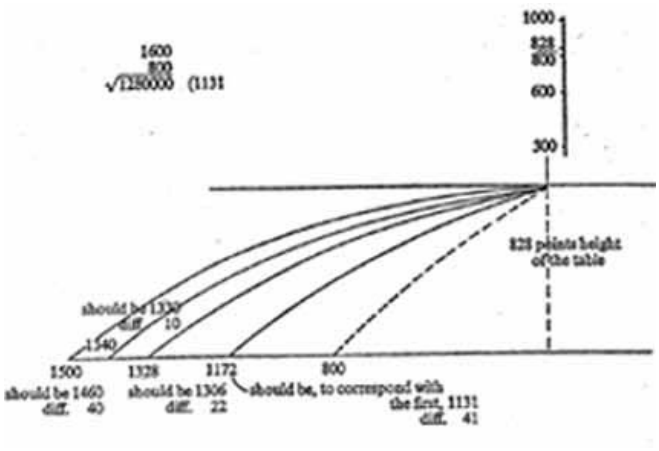


Figure 3: Galileo's projectile motion graph. The graph combines constant horizontal velocity and uniform vertical acceleration, yielding an equation of $y = (g/2v_x^2)x^2$. It is important to note that Galileo conducted his experiment from an elevated position.
Image from The Galileo Project

Our lab results fairly match Galileo's graph of a smooth parabola with two independent components. The maximum horizontal distance of the marble shooter was reached at 45°. We observed that the marble's horizontal velocity remained relatively constant and since each marble angle was tested for 5 trials each, we were able to achieve greater accuracy in our representation of the marble's trajectory.

Sources of Experimental Errors:

The launch angle was measured using a plumb line on the included protractor. Between 0° and 82.5°, there was one outlier which corresponds to about 2.6 in from its expected value. We did not measure launch height, so our fitted value 3.6 ± 0.5 in is an estimate. Air resistance (which the equations do not take into account) tends to increase the distance traveled on the shortest and longest flights, bringing the angle with the maximum distance traveled below 45°. The shooter also varied from shot to shot, at most 30° (s.d. 1.41°) to 52.5° (s.d. 1.06°). Additionally, the marble's bouncing or rolling after making initial contact may have smeared the carbon-paper marks. Lastly, five trials per angle is a small number, so the 37.5/52.5 result shows statistical significance prior to the Bonferroni correction, indicating room for greater accuracy.

Potential follow-up experiments could film the flight at 120 fps or higher to track the marble and recover the entire flight path. They could also measure the launch height, employ a digital angle gauge, increase the number of shots per launch angle, and repeat at more powerful spring levels to see how drag increases with speed.

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Thank you, Mrs. J. Komitas and the Science and Engineering Magnet Program, for providing us with the resources and lab space to conduct our research. Lucy Gulley carried out the experiment, recorded distances, wrote the discussion and acknowledgements. Bheshaja Koduri wrote the methodology, sources of experimental error, and did the statistical analysis. Hailey Perle also carried out the experiment, and wrote the abstract. Neha Selvadhurai graphed data, recorded results, and researched alternative methods of deriving projectile motion.

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Angles	# of Shots	Distance Traveled (inches)	Angles	# of Shots	Distance Traveled (inches)
		9			62.81
		9.5			63.38
		9			62.81
		9.5			62.38
0	5	7.5	45	5	62.56
		24.25			62.7
		24.15			60.75
		24.81			60
		24.44			60.38
7.5	5	22.88	52.5	5	60.56
		38.31			52.63
		38.43			52.13
		38.5			52.88
		38.75			52.69
15	5	38.38	60	5	51.5
		50.31			41.81
		50.06			40
		50.13			40.06
		50.44			40.81
22.5	5	50.31	67.5	5	40.69
		60.81			27.75
		58			26.81
		57.56			27.81
		57.38			27.5
30	5	57.93	75	5	27.88
		62.5			12
		62.44			10.81
		62.31			11.13
37.5	5		82.5	5	

		62.38			11.56
		62.25			11.44